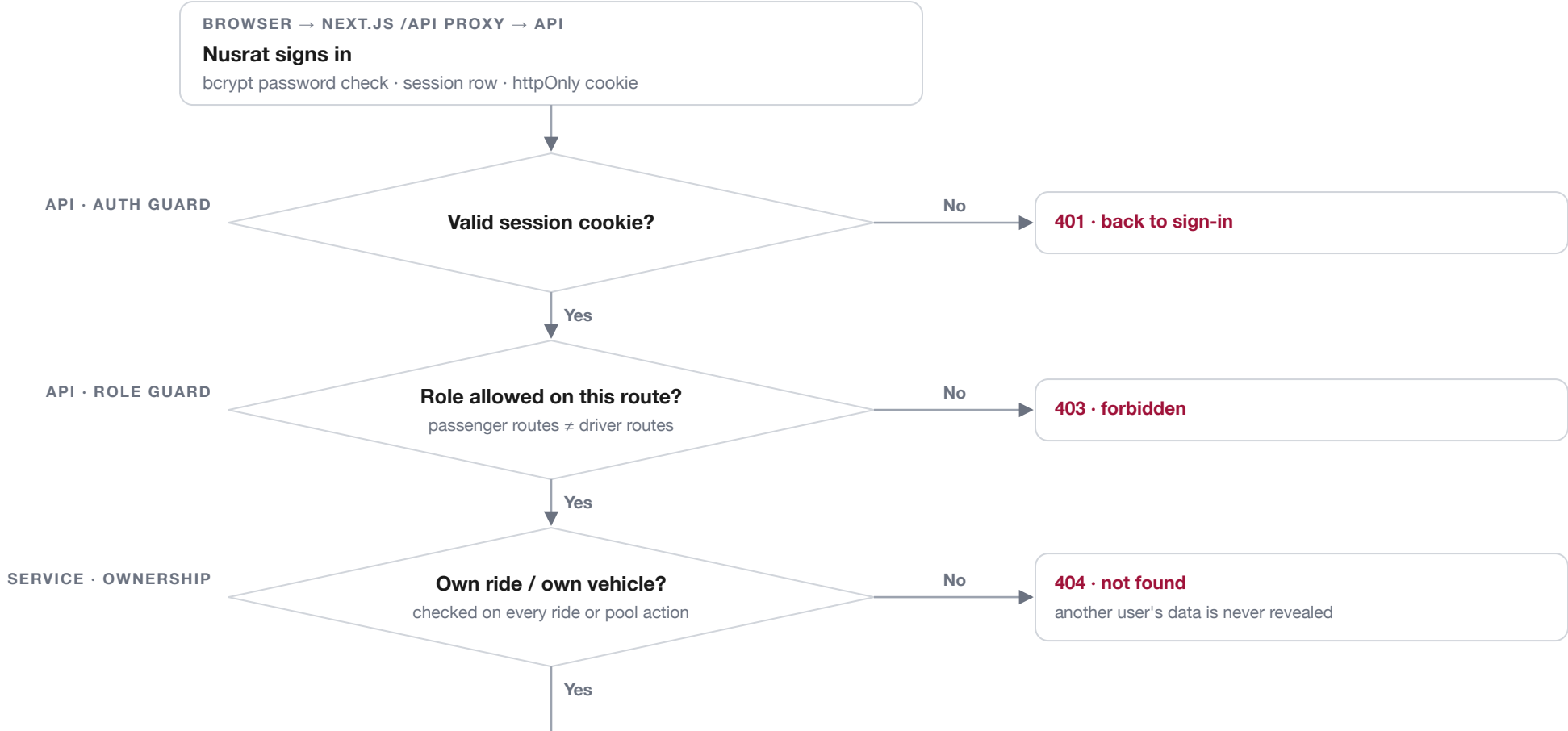


Dhaka Tesla Pool · Ride Flow

How one ride moves through the system · read top to bottom · side notes cover concurrent and edge cases

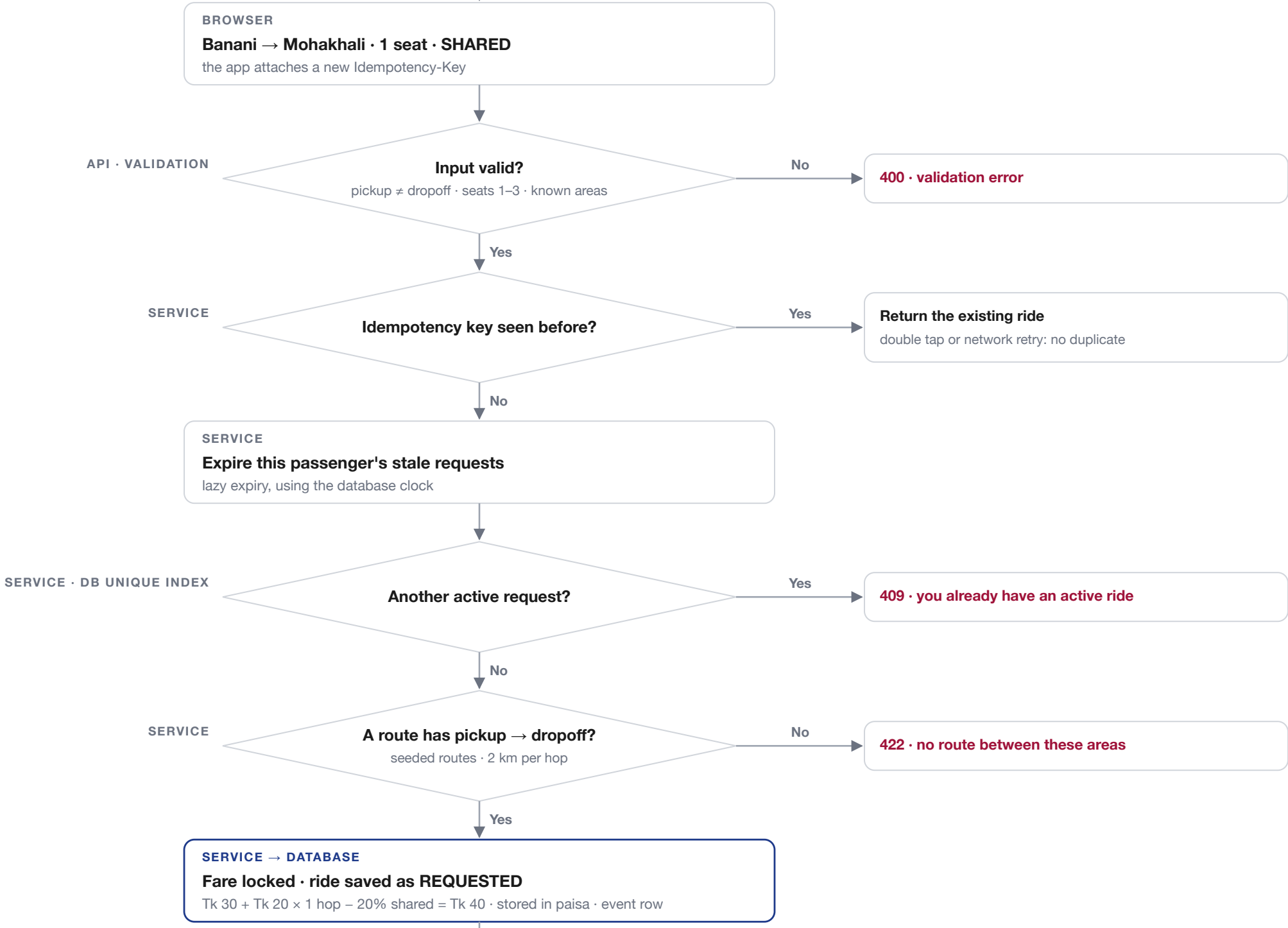


1 SIGN IN



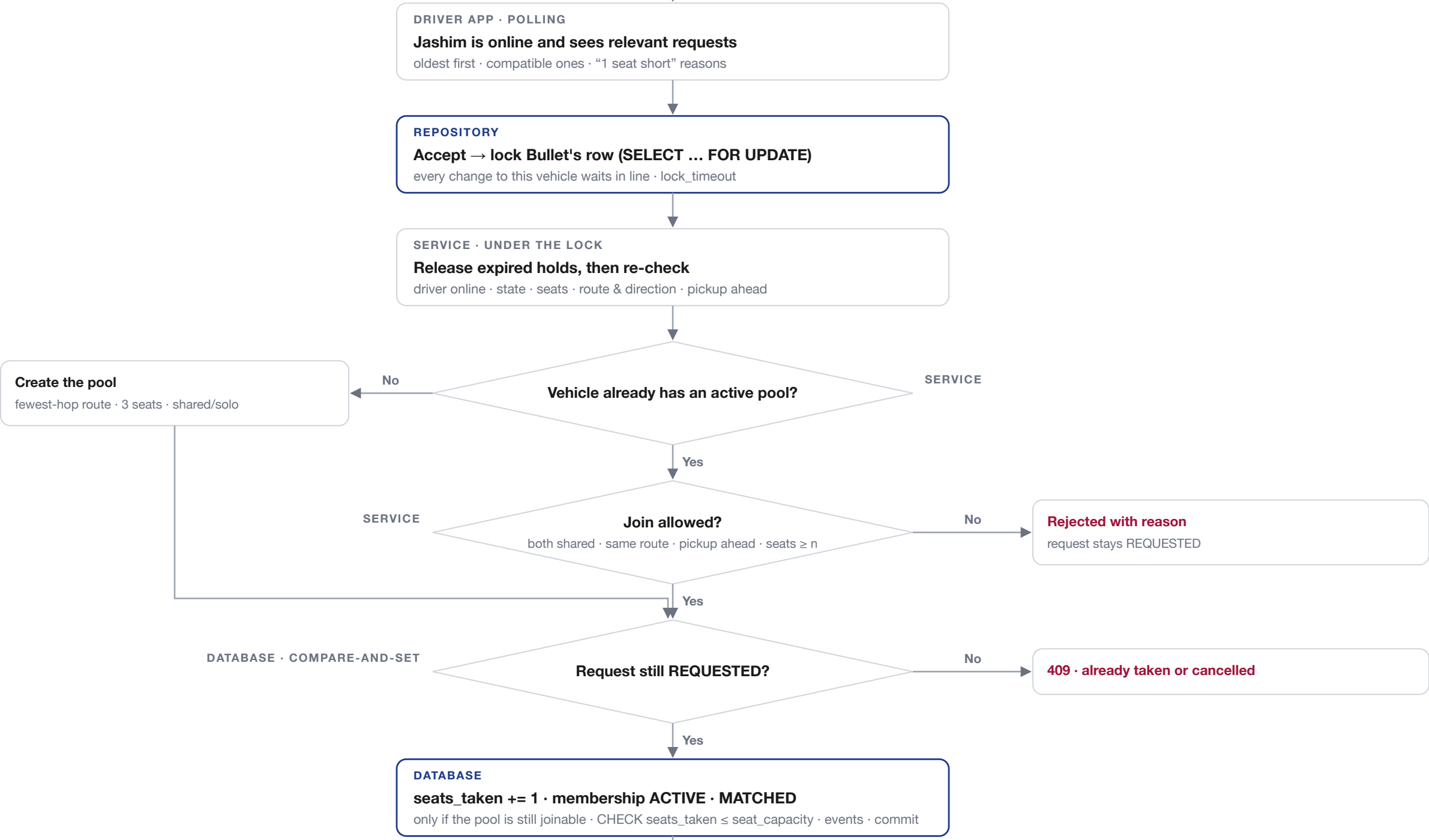
Sign up
Unique email (case-insensitive)
Password hashed with bcrypt
Login attempts rate-limited
Session expires after 12 h

2 REQUEST A RIDE



Double tap (idempotency)
Same key → same ride returned
Two inserts at once: any unique error → look up by key first

3 MATCHING



Fare: shared vs solo
SHARED: Tk 30 + Tk 20 × hops – 20%
SOLO: no discount · empty vehicle only
Locked at request, never changes
Too few seats → "N seat(s) short"

Seat hold (planned)
New SHARED request fits an active pool → seat HELD 60 s
Driver confirms → MATCHED
Reject / timeout → REQUESTED

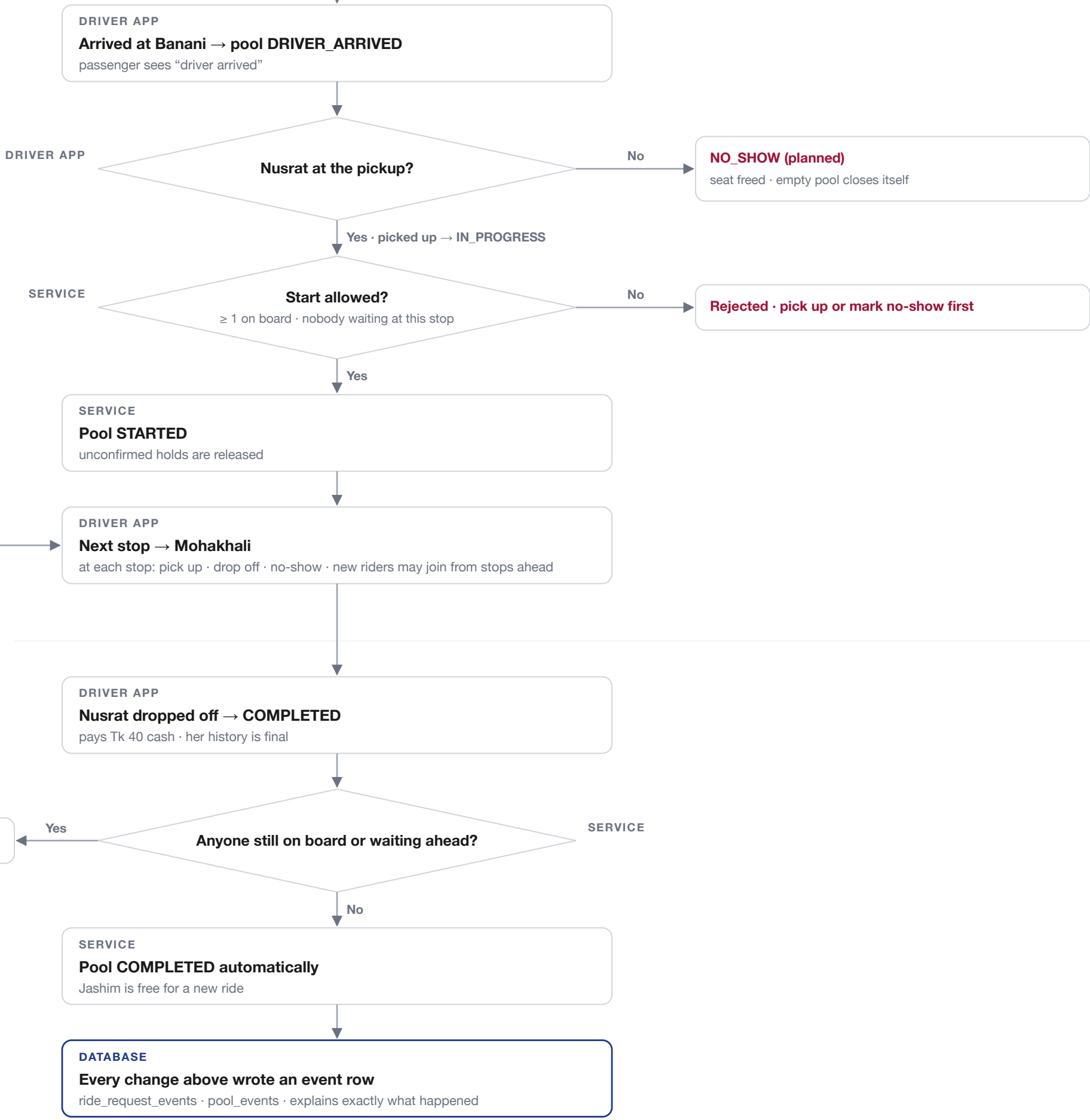
Last seat, two people at once
Rafiq & Shirin both want seat 3
Both wait for Bullet's lock
First: gets it · second: 0 free
→ stays REQUESTED, not dropped

No pool yet, two accepts at once
First accept creates the pool
Second waits, then joins it
One pool, two passengers

Retries and timeouts
Lock wait > lock_timeout → retryable error, nothing changed
Repeated driver action returns the current state (idempotent)

Database guarantees
seats_taken ≤ seat_capacity
1 active pool per vehicle
1 active membership per request
1 active request per passenger

4 THE TRIP



Online / offline
Changed under the vehicle lock
Offline refused while a pool is active · accept checks online

What the passenger sees
Polls her ride every few seconds
Own status and own fare only
Driver + vehicle name
Co-riders only as a count

Cancellation rules
Passenger: until picked up
→ seat freed immediately
Driver: before start → riders back to REQUESTED, fresh expiry
Empty before start → pool closes
After pickup: no cancel

Request expiry (planned)
Nobody accepts in 5 min
→ EXPIRED · checked lazily under the lock

5 END OF RIDE